

ALERT BAY, BC, Canada

WMO: 741100

Lat: 50.582N

Lon: 126.915W

Elev: 59

StdP: 100.61

Time Zone: -8.00 (NAP)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.1	-10.0	1.6	-2.4	-6.9	2.1	-1.4	26.3	6.4	25.5	5.8	4.3	140	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	20.3	15.5	18.6	14.6	17.3	13.9	16.1	18.9	15.2	17.7	14.4	16.7	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.6	17.5	14.0	10.1	16.2	13.3	9.6	15.3	45.2	18.9	42.5	17.7	40.4	16.9	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.1	15.9	-5.9	26.3	3.6	1.2	-8.4	27.2	-10.5	27.9	-12.5	28.6	-15.1	29.5
			-7.2	19.3	3.7	1.4	-9.8	20.3	-12.0	21.1	-14.1	21.9	-16.7	22.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	4.1	4.5	6.2	7.9	10.4	12.0	13.9	14.4	12.1	9.4	6.2	4.3
	DBStd	4.23	2.50	3.63	2.10	2.05	1.96	1.50	1.39	1.50	1.85	2.37	2.10	2.63
	HDD10.0	880	183	155	119	69	18	2	0	0	2	39	116	178
	HDD18.3	3482	441	387	377	313	247	191	137	123	187	277	366	436
	CDD10.0	441	0	1	1	7	29	61	121	136	66	20	1	0
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH23.3	13	0	0	0	0	1	2	2	3	5	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.7	7.7	6.3	5.1	3.9	3.6	3.0	3.0	2.5	2.5	4.8	7.1	6.9
Precipitation	PrecAvg	1668	224	154	128	99	72	71	55	67	114	227	235	222
	PrecMax	2195	458	262	228	176	146	172	150	154	229	461	675	404
	PrecMin	1080	70	44	30	33	23	21	7	1	17	68	81	91
	PrecStd	254	95	63	51	35	34	36	38	40	55	94	119	75
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	12.8	13.6	17.9	20.6	19.7	22.2	22.9	23.0	18.2	11.7	9.9
		MCWB	7.5	9.1	8.8	10.6	14.5	15.1	16.5	16.9	14.2	14.2	10.6	8.0
	2%	DB	8.7	10.9	11.7	14.7	17.1	17.2	19.1	20.1	18.1	15.6	10.8	8.8
		MCWB	7.3	7.7	7.1	9.8	12.4	13.8	14.8	15.8	14.6	12.7	9.8	7.5
	5%	DB	8.0	9.7	10.4	12.7	15.3	15.9	17.8	18.5	16.3	13.9	9.8	8.2
		MCWB	6.7	7.3	7.3	8.9	11.7	12.6	14.2	14.9	13.4	11.8	8.8	7.1
	10%	DB	7.2	8.8	9.3	11.2	13.9	14.9	16.7	17.3	15.1	12.6	9.0	7.6
		MCWB	5.9	6.9	6.9	8.4	11.0	12.1	13.7	14.3	12.7	11.0	8.0	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	9.8	10.1	11.9	14.7	15.9	17.2	17.6	16.2	15.1	11.2	8.3
		MCDB	9.3	12.6	11.1	16.9	20.0	18.2	20.2	21.8	19.4	17.2	11.8	9.0
	2%	WB	7.6	8.4	8.6	10.4	13.2	14.3	15.5	16.2	14.9	13.3	10.0	7.8
		MCDB	8.4	10.0	10.2	13.5	16.2	16.8	18.4	19.1	17.0	14.9	10.6	8.6
	5%	WB	6.7	7.7	8.0	9.6	12.2	13.3	14.6	15.5	13.9	12.0	9.0	7.3
		MCDB	7.6	9.1	9.4	11.9	14.4	15.1	17.1	17.8	15.9	13.3	9.5	8.1
	10%	WB	6.1	7.1	7.3	8.7	11.4	12.6	13.9	14.7	13.1	11.2	8.1	6.7
		MCDB	7.1	8.4	8.7	11.0	13.7	14.3	16.1	16.7	14.7	12.5	8.8	7.5
Mean Daily Temperature Range	5% DB	MDBR	2.8	4.0	5.1	5.8	6.2	5.6	5.8	6.2	6.1	4.5	2.8	2.6
		MCDBR	3.2	5.0	7.3	9.5	9.5	7.6	8.0	8.8	8.4	7.0	3.2	2.7
		MCWBR	2.5	3.1	4.3	5.3	5.6	4.6	4.7	5.0	5.2	4.6	2.5	2.1
	5% WB	MCDBR	2.9	4.0	5.0	7.7	8.4	6.8	7.3	7.9	7.9	6.4	3.1	2.4
		MCWBR	2.5	2.8	3.5	4.7	5.3	4.5	4.7	4.9	5.0	4.4	2.6	2.1
Clear-Sky Solar Irradiance	taub	0.316	0.319	0.348	0.361	0.373	0.367	0.369	0.358	0.353	0.362	0.323	0.307	
	taud	2.423	2.497	2.411	2.399	2.397	2.450	2.480	2.527	2.524	2.390	2.442	2.403	
	Ebn at Noon	717	825	855	882	883	890	882	877	843	749	700	664	
	Edh at Noon	62	73	95	107	112	107	103	93	85	81	60	54	
All-Sky Solar Radiation	RadAvg	0.75	1.62	2.46	3.79	4.91	5.14	5.47	4.75	3.31	1.76	0.87	0.60	
	RadStd	0.10	0.27	0.34	0.31	0.55	0.56	0.68	0.41	0.31	0.29	0.15	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page